

Chapter 4

32 What is Scaling? Explain horizontal and vertical scaling with examples.

→ Scaling refers to the process of adjusting computing resources to handle changes in workload or demand in a cloud environment. It ensures that applications remain responsive and performant even during traffic spikes.

Cloud scaling can be done in two ways:
Horizontal scaling and Vertical scaling.

① Horizontal Scaling

- It involves adding or removing multiple instances of servers or resources to handle increased or decreased load. It distributes the workload across multiple nodes.

Features

- Improves availability and fault tolerance
- Supports cloud elasticity.
- often used with load balancers.

Example:

- Adding more virtual machines to a web server cluster in AWS EC2 when user traffic increases.
- Removing instances when traffic decreases.

2 Vertical Scaling

It involves increasing or decreasing the capacity of a single server by adding more CPU, RAM, or storage features.

- Simple to implement than horizontal scaling.
- No need to modify application architecture.

Example:

- Upgrading an AWS RDS instance from 4 vCPUs to 8 vCPUs to handle more database queries.
- Downgrading when demand is low.

Q 33 What is auto-scaling? Provide a step-by-step guide to configure auto-scaling in a cloud environment.

- Auto-scaling is a cloud feature that automatically adjusts computing resources based on current demand. It ensures optimal performance and cost-efficiency by adding or removing instances dynamically when workload changes. Auto-scaling is commonly used with applications that experience variable traffic.

Step-by-Step Guide to configure Auto-Scaling in a Cloud Environment

Step 1: Define Launch Configuration or Template

- Creates a launch configuration / template specifying:
 - AMI (Amazon Machine Image)
 - Instance type (CPU, memory)
 - Security groups
 - Key pair for SSH access

Step 2: Create an Auto-Scaling Group (ASG)

- Specify the VPC and subnets where instances will be launched.
- Define minimum, maximum, and desired number of instances
- Associate the launch configuration / template.

Step 3: Configure Scaling Policies

- Dynamic Scaling: Add rules to scale out (add instances) or scale in (remove instances) based on metrics such as CPU usage, network, network traffic, or custom metrics.
- Scheduled ~~str~~ Scaling: Set specific times to scale resources up or down.

Step 4: Attach Load Balancer

- Integrate with load balancer to distribute traffic across multiple instances
- Ensures availability and efficient utilization.

of resources.

Step 5: Monitoring and Notifications

- Enable CloudWatch or similar monitoring service to track performance metrics.
- Configure notifications (SNS or email) for scaling events.

Step 6: Test Auto-Scaling

- Simulate increased traffic to ensure new instances launch automatically.
- Simulate reduced load to check that unused instances terminate.

Q 34 Public cloud gives high scalability but low control. Discuss this trade-off logically.

→ Public Clouds can instantly allocate resources such as virtual machines, storage, or databases based on demand.

→ Public Cloud platforms, such as AWS, Azure and Google Cloud, provide computing resource over the internet on a shared infrastructure. They are designed to be highly scalable, but this comes with a trade-off in control.

High Scalability in Public Cloud

- Public Clouds can instantly allocate resources such as virtual machines, storage, or

databases based on demand.

- Users can scale applications up or down dynamically without worrying about physical hardware.
- This elasticity is ideal for variable workloads, like e-commerce websites, streaming services, or analytics platforms.

Low Control in Public Cloud.

- Users do not manage the underlying infrastructure, including hardware, networking or hypervisors.
- Customization is limited to the software stack and resource configurations provided by the cloud provider.

Logical Trade-off

1. Scalability vs Control

- High scalability is achieved because the cloud provider manages infrastructure at a massive scale.
- Low control is a natural consequence since users cannot intervene in underlying hardware or virtualization layers.

2. Cost Efficiency vs Customization:

- Users save costs and operational effort by leveraging shared infrastructure.
- Full control, like in private clouds, would

require more investment in hardware and IT staff.

3. Use Case Alignment:

- Public clouds are best for applications needing rapid scaling with minimal management overhead.
- Applications requiring full control over infrastructure, security, or regulatory compliance might prefer private clouds.

Q35 Explain load balancing and its role in cloud resources distribution.

⇒ Load Balancing is the process of distributing incoming network traffic or workloads across multiple servers or computing resources to ensure that no single server is overloaded. It is a fundamental mechanism used in cloud computing to improve application performance, availability, and reliability.

A load balancer acts as an intermediary between users and backend servers. When a user sends a request, the load balancer forwards it to one of the available servers based on predefined algorithms such as round robin, least connections, or resource-based routing. If a server fails, traffic is redirected to healthy servers automatically.

Role of Load Balancing in Cloud Resource Distribution

1, Efficient Resource Utilization

Load balancing ensures that workloads are evenly distributed across cloud resources. This prevents some servers from being overutilized while others remain idle, leading to better overall resource usage.

2, High Availability

By distributing traffic across multiple instances, load balancing eliminates single points of failure. If one server becomes unavailable, the load balancer routes requests to other operational servers, ensuring continuous service.

3, Scalability

Load balancers work closely with auto-scaling mechanisms. When demand increases, new instances are added and automatically included in the load-balancing pool, enabling seamless horizontal scaling.

4, Improved Performance

Load balancing reduces response time by directing requests to the least-loaded or closest server. This results in faster processing and a better user experience.

5, Fault Tolerance

Load balancers continuously monitor server health. Failed or unhealthy instances are removed from traffic distribution, enhancing system reliability.

Q36 Describe redundancy and high availability systems
⇒ Redundancy and high availability (HA) are core design principles in cloud computing and distributed systems that ensure continuous service operation even in the presence of failures.

Redundancy refers to the duplication of critical system components such as servers, storage, network links, or data. The purpose is to eliminate single points of failure.

- Data is often replicated across multiple disks, servers or data centers.

- Hardware redundancy includes backup servers, power supplies, and network paths.

- If one component fails, another redundant component takes over without data loss.

Example: Storing multiple copies of data across different availability zones in a cloud platform.

High Availability ensures that systems remain operational for most of the time, typically expressed as a percentage such as 99.9% or 99.99% uptime.

- HA systems use redundancy combined with automatic failover mechanisms.
- Monitoring tools detect failures and redirect traffic to healthy resources.
- Load balancers and clustered systems are commonly used to achieve HA.

Example:

A web application running on multiple cloud instance behind a load balancer.

Redundancy provides the backup components, while high availability defines how quickly and smoothly the system recovers from failures. Redundancy is a requirement, but HA also needs automation, monitoring and failover strategies.

Q37 Discuss Service Level Agreements and their types.

→ A Service level Agreements (SLA) is a formal contract between a cloud service provider and a customer that defines the expected level of service. It clearly specifies service performance standards, responsibilities, and remedies in case the agreed service levels are not met.

Key Components of an SLA

- Service scope and description
- Performance metrics
- Availability guarantees
- Security and Compliance requirements
- Penalties or service credit

Types of Service Level Agreements

1. Customer-Based SLA

A customer-based SLA is designed for a specific customer and covers all services provided to that customer. It is customized according to the customer's requirements.

Example

An SLA between a cloud provider and a single enterprise client covering compute, storage and support services

2. Service-Based SLA

A service-based SLA applies to all customers using a particular service. The same service-level terms are offered uniformly

Example

An SLA guaranteeing 99.9% uptime for all users of a cloud storage service.

3, Multi-level SLA

A multi-level SLA is structured into multiple layers to address different levels of ~~end~~ service. It typically includes corporate-level, customer-level and service-level agreements.

Example: - Corporate SLA: general policies and support hours
- Customer SLA: customer-specific requirements
- Service SLA: technical metrics for individual services.

Q38 Explain billing management systems used in cloud computing.

→ A billing management system in cloud computing is used to measure, track and charge users based on their actual consumption of cloud resources. It supports the pay-as-you-use model, which is a core feature of cloud services.

Components:

• Usage Metering:

- Monitors actual resource consumption.

• Pricing Engine

- Applies rates to usage data.

• Billing & Invoicing Module:

- Generates bills automatically.

- Date _____
Page _____
- Customer Portal :
 - provides cost visibility and reports :

Functions of Billing Management Systems

1. Usage Metering

The System continuously monitors resource usage such as CPU hours, storage consumed, network bandwidth, and number of API requests. Accurate metering ensures fair and transparent billing.

2) Cost-Calculation

Based on predefined pricing models, the billing system calculates charges for each resource used. Pricing may vary by service type, region, or usage duration.

3) Invoice Generation

Billing systems generate detailed invoices showing resource usage, costs, taxes, and discounts. Invoices are usually produced monthly and are accessible through cloud dashboards.

4) Account and Subscription Management

Users can manage accounts, service plans, budgets, and payment methods. This allows organizations to track spending across projects or departments.

Types of Cloud Billing Model

- Pay-as-you-go : Users pay only for resources consumed.
- Subscription-based :- Fixed recurring charges
- Reserved or committed usage :- Discounts for long-term commitments.

Chapter - 5

Q39 What are cloud security threats? Discuss major modern attacks.

⇒ Cloud Security threats are potential risks that can compromise the confidentiality, integrity and availability of data and services hosted in cloud environments. Due to shared infrastructure, internet accessibility, and complex configurations, cloud systems face various modern security challenges.

Major Modern Cloud Security Attacks

1, Data Breaches.

Data breaches occur when unauthorized users gain access to sensitive cloud data such as personal information, financial records, or intellectual property. Misconfigured storage services and weak access controls are common cause

2, Account Hijacking

Account Hijacking happens when attackers steal login credentials through phishing, malware, or brute force attacks. Once compromised, attackers can manipulate cloud resources, steal data, or disrupt services. Weak passwords and lack of multi-factor authentication increase this risk.

3) Denial of Service (DoS and DDoS) Attacks

In DoS and distributed Denial of Service (DDoS) attacks, attackers flood cloud services with massive traffic, making them unavailable to legitimate users. Although cloud providers offer DDoS protection, large-scale attacks can still impact performance and availability.

4) Insecure APIs and Interfaces

Cloud services rely heavily on APIs for management and integration. Poorly secured APIs can be exploited to gain unauthorized access or manipulate cloud resources. Lack of encryption and weak authentication are common vulnerabilities.

5) Misconfiguration Attacks

Misconfigurations such as open storage buckets, overly permissive security rules, or exposed ports are major causes of cloud attacks. Attackers exploit these errors to access data or deploy malicious code.

Q40 Explain Security as a Service and its architecture.

→ Security as a Service (SECaaS) is a cloud-based service model in which security functions are delivered on demand over the internet.

Q41 Write the importance of data security in the cloud. Explain with one real cloud data breach example.

⇒ Data security is a critical concern in cloud computing because sensitive information such as personal data, financial records, and business data is stored and processed on remote servers accessed over the internet. Ensuring data security helps data from unauthorized access, data loss, and cyberattacks.

Firstly, strong data security maintains Confidentiality by ensuring that only authorized users can access cloud data. Techniques such as encryption, identity and access management, and multi-factor authentication are used to protect sensitive information. This is especially important when data is shared across multiple users and locations.

Secondly, data security ensures data integrity, meaning the data remains accurate and unaltered. Security controls prevent attackers or unauthorized users from modifying or corrupting data. This is essential for maintaining trust in cloud-based applications and services.

Strong data security helps organizations meet legal and regulatory requirements such as data protection laws. Failure to secure data can result in legal penalties, financial loss, and reputational damage.

Real Cloud Data Breach Example: Capital One (2019)

In 2019, Capital One suffered a major cloud data breach where an attacker gained unauthorized access to data stored on Amazon Web Services (AWS). The breach exposed personal information of over 100 million customers, including names, addresses, credit scores, and bank account details.

The attack was caused by misconfigured firewall and improper access controls, which allowed the attacker to exploit a vulnerability in the cloud environment. This incident highlighted that while cloud providers offer secure infrastructure, customers are responsible for correctly configuring security settings under the shared responsibility model.

Q42 Describe cloud security dimensions

→ Cloud security dimensions represent the key aspects that must be protected to ensure safe and reliable cloud computing. These dimensions focus on safeguarding data, applications, and infrastructure while maintaining trust, compliance and service continuity:

1. Confidentiality

Confidentiality ensures that cloud data is accessible only to authorized users. It is maintained using

encryption, identity and access management (IAM), and authentication mechanisms. Protecting confidentiality prevents data leaks and unauthorized access.

2. Integrity

Integrity ensures that data remains accurate and unaltered during storage and transmission. Techniques such as hashing, digital signature and access controls are used to prevent unauthorized modification. This ensures trust in cloud data and applications.

3. Availability

Availability ensures that cloud services and data are accessible whenever required. Redundancy, fault tolerance, backups, and disaster recovery mechanisms are used to prevent downtime. High availability is critical for business continuity.

4. Authentication and Authorization

Authentication verifies the identity of users, while authorization defines what resource they can access. Strong authentication methods like multi-factor authentication (MFA) enhance cloud security. Proper authorization minimizes insider and external threats.

5 Data & Privacy

Data Privacy focuses on protecting personal and sensitive information. It ensures compliance with data protection laws and regulations. Privacy controls prevent misuse or unauthorized sharing of user data.

6 Compliance and Governance

Compliance ensures that cloud systems follow legal, regulatory, and organizational policies. Governance frameworks define rules for data handling, auditing and risk management. This dimension is crucial for regulated industries.

Q43 Explain IAM, Hashing and Digital Signature in cloud.

⇒ In cloud computing, Id

→ Cloud security relies on multiple mechanisms to protect users, data and services. Identity and Access Management (IAM), hashing and digital signatures play a vital role in ensuring secure cloud operations.

1. Identity and Access Management (IAM)

IAM is a foundational security framework that defines and manages the roles and access privileges of individuals and non-human entities (like applications or services) within a cloud environment.

- **Authentication** :- Verifies that a user is who they claim to be, typically using methods like passwords, multi-factor authentication (MFA) or biometric ~~sen~~ scans.
- **Authorization** :- Determines what an authenticated user is allowed to do and which resources they can access. This is often managed through policies and role-based access control (RBAC) following the principle of "least privilege" - users get only the permissions necessary to perform their tasks.
- **Centralized Control** :- IAM solutions, such as AWS IAM, Azure AD (Microsoft Entra ID), and Google Cloud IAM, provide a central platform for managing permissions across an organization's entire cloud infrastructure.

Hashing

Hashing is a mathematical process that transforms input data of any size into a fixed-length string of characters called a hash value or message digest.

- **One-Way function** :- Hashing is irreversible; the original data cannot be reconstructed from the hash alone, making it ideal for securely storing sensitive information like passwords.

- **Data Integrity:** A minor change in the input data results in a completely different hash value. This property ensures data has not been altered during transmission or storage.

- **Examples:** Common algorithms includes SHA-256 and SHA-3, which are considered strong and collision-resistant (it's nearly impossible for two different inputs to produce the same hash).

Digital Signatures

A digital signature is a cryptographic technique used to validate the authenticity and integrity of a digital message or document, leveraging hashing and public key cryptography.

Process:

1. The sender calculates a hash of the document (the "digital fingerprint").
2. The sender then encrypts this hash using their private key. The encrypted hash is the digital signature.
3. The recipient uses the sender's publicly shared public key to decrypt the signature and obtain the original hash.
4. The recipient independently calculates a new hash of the received document and compares it to the decrypted hash.

In cloud computing, digital signatures are used for secure authentication of API requests and ensuring the legal validity of electronic documents processed through cloud services like Adobe Sign or DocuSign.

Q44 What is disaster management in cloud computing? Explain methods.

→ Disaster management in cloud computing refers to the strategies and processes used to prepare for, respond to and recover from unexpected events such as hardware failures, cyberattacks, natural disasters, or data center outages. The main goal is to ensure business continuity, minimize data loss, and reduce downtime.

Cloud disaster management strategies fall along a spectrum of cost and recovery speed, allowing organizations to choose the approach that best fits their specific RTO and RPO requirements for different applications.

1. Backup and Restore

This is the most cost-effective approach, suitable for less critical data that can tolerate longer downtime.

- Method: Data and application snapshots are regularly backed up to the cloud. In a disaster,

, the systems are restored from the most recent backup.

- Use Case :- Archiving data for compliance or non-critical data that is not frequently updated.

2. Pilot Light (Warm Disaster Recovery)

This method involves running a minimal, "pilot light" version of the core application in the cloud, while continuously replicating critical data.

- Method :- The core services are always running but at a minimal capacity. In a disaster, the standby environment is scaled up to full capacity and live traffic is redirected to it.

- Use Case : Applications requiring faster recovery than backup and restore but where the cost of a full, active replica is not justified.

3. Warm Standby

This strategy involves maintaining a scaled-down but fully functional ~~group~~ copy of the entire production environment in the cloud.

- Method : The duplicate environment runs continuously but under a reduced load. Failover involves a quick process of scaling up the resources and redirecting users, resulting in relatively fast recovery times.

Q45, Even with encryption and IAM, cloud breaches still happen. Provide local reasons.

→ Even though encryption and Identity and Access Management (IAM) are essential cloud security controls, they alone cannot guarantee complete protection. Cloud breaches still occur due to several logical and practical reasons.

1) Misconfiguration of Cloud Resources.

Many breaches happen because encryption or IAM policies are incorrectly configured. Examples include publicly exposed storage buckets or overly permissive access roles. Human error remains a major security risk in cloud environments.

2. Stolen or Compromised Credentials.

Encryption does not protect data if attackers gain access using valid credentials. Phishing attacks, weak passwords, and lack of multi-factor authentication allow attackers to bypass IAM controls.

3. Insider Threats

Authorized users such as employees or contractors may misuse their access, intentionally or accidentally. IAM grants access but cannot always prevent malicious insider actions.

4. Application-Level Vulnerabilities

Encryption protects data ~~are~~ at rest and in transit, but vulnerabilities in applications (such as SQL injection or insecure APIs) can still be exploited. Attackers may access decrypted data during processing.

5. Poor Key Management

If encryption keys are poorly protected or stored insecurely, attackers can decrypt sensitive data. Weak key rotation and improper key access policies increase this risk.